

Eenheid 6.4

Note Title

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Die kruisprodukt

Stewart p 814

Die kruisprodukt wordt mbv determinante bepaald.

Definitie: Laat $a, b, c, d \in \mathbb{R}$

in Tweede orde determinant is

$$* \begin{vmatrix} a & b \\ c & d \end{vmatrix} = ad - bc$$

Voorbeelden: ① $\begin{vmatrix} -1 & 2 \\ 3 & 4 \end{vmatrix} = (-4) - 6 = -10$ ② $\begin{vmatrix} 3 & 4 \\ -1 & 2 \end{vmatrix} = 6 - (-4) = 10$

Opmerking: Die kruisprodukt $\bar{a} \times \bar{b}$ gee 'n vektor wat loodreg is op beide $\bar{a}$ en $\bar{b}$.
Dan is $(\bar{a} \times \bar{b}) \cdot \bar{a} = 0$ en $(\bar{a} \times \bar{b}) \cdot \bar{b} = 0$

Definitie van kruisproduct. ($\bar{a} \times \bar{b} = \text{vektor}$)

Laat $\bar{a} = \langle a_1, a_2, a_3 \rangle$ en $\bar{b} = \langle b_1, b_2, b_3 \rangle$

$$\text{dan is } \underline{\bar{a} \times \bar{b}} = \begin{vmatrix} \bar{i} & \bar{j} & \bar{k} \\ a_1 & a_2 & a_3 \\ b_1 & b_2 & b_3 \end{vmatrix}$$

$$\begin{array}{l} \bar{a} \neq \bar{0} \\ \bar{b} \neq \bar{0} \\ \bar{a}, \bar{b} \in \mathbb{R}^3 \end{array}$$

$$= \bar{i} \begin{vmatrix} a_2 & a_3 \\ b_2 & b_3 \end{vmatrix} - \bar{j} \begin{vmatrix} a_1 & a_3 \\ b_1 & b_3 \end{vmatrix} + \bar{k} \begin{vmatrix} a_1 & a_2 \\ b_1 & b_2 \end{vmatrix}$$

$$= \bar{i} (a_2 b_3 - a_3 b_2) - \bar{j} (a_1 b_3 - a_3 b_1) + \bar{k} (a_1 b_2 - a_2 b_1)$$

VBD: Bepaal $\bar{u} \times \bar{v}$ als $\bar{u} = \langle -3, 4, -2 \rangle$ en $\bar{v} = \langle 2, -1, 3 \rangle$. Toon aan dat you antw korrek is (wys dat $(\bar{a} \times \bar{b}) \cdot \bar{a} = 0$ en $(\bar{a} \times \bar{b}) \cdot \bar{b} = 0$)

$$\underline{\underline{\bar{u} \times \bar{v}}} = \begin{vmatrix} \bar{i} & \bar{j} & \bar{k} \\ -3 & 4 & -2 \\ 2 & -1 & 3 \end{vmatrix}$$

$$\begin{aligned} &= \bar{i} \begin{vmatrix} 4 & -2 \\ -1 & 3 \end{vmatrix} - \bar{j} \begin{vmatrix} -3 & -2 \\ 2 & 3 \end{vmatrix} + \bar{k} \begin{vmatrix} -3 & 4 \\ 2 & -1 \end{vmatrix} \\ &= (12 - 2)\bar{i} - (-9 + 4)\bar{j} + (3 - 8)\bar{k} \\ &= 10\bar{i} + 5\bar{j} - 5\bar{k} = \langle 10, 5, -5 \rangle \end{aligned}$$

ONTWOU : $\bar{u} \times \bar{v} = -(\bar{v} \times \bar{u}) = -\langle 10, 5, -5 \rangle = \langle -10, -5, 5 \rangle$

Op somming

• Laat $\bar{a}, \bar{b} \in \mathbb{R}^3$, $\bar{a} \neq \bar{0}$, $\bar{b} \neq \bar{0}$, dan

1) $\bar{a} \times \bar{b}$ is loodreg / ortogonaal op beide $\bar{a}$ en $\bar{b}$

2) $\bar{a}$ en $\bar{b}$ is parallel aa $\bar{a} \times \bar{b} = \bar{0} = \langle 0, 0, 0 \rangle$

* 3) $|\bar{a} \times \bar{b}|$ is die oppervlakte van die parallelogram wat deur $\bar{a}$ en $\bar{b}$ bepaal word.

Eienskappe:

Laat $\bar{a}$, $\bar{b}$ en $\bar{c}$ vektore in $\mathbb{R}^3$ wees en $k \in \mathbb{R}$

* 1. $\bar{a} \times \bar{b} = -(\bar{b} \times \bar{a})$

Skalaar

2. $(k\bar{a}) \times \bar{b} = k(\bar{a} \times \bar{b})$

$\bar{a} \times (k\bar{b}) = k(\bar{a} \times \bar{b})$

3. $\bar{a} \times (\bar{b} + \bar{c}) = \bar{a} \times \bar{b} + \bar{a} \times \bar{c}$

4. $(\bar{a} + \bar{b}) \times \bar{c} = \bar{a} \times \bar{c} + \bar{b} \times \bar{c}$

Voorbeeld: ① Laat $\bar{a} = \langle 1, 2, 0 \rangle$

$\bar{b} = \langle 3, -1, 2 \rangle$

$$\bar{a} \times \bar{b} = \rightarrow \begin{vmatrix} \bar{i} & \bar{j} & \bar{k} \\ 1 & 2 & 0 \\ 3 & -1 & 2 \end{vmatrix}$$

$$= \begin{vmatrix} 2 & 0 \\ -1 & 2 \end{vmatrix} \bar{i} - \begin{vmatrix} 1 & 0 \\ 3 & 2 \end{vmatrix} \bar{j} + \begin{vmatrix} 1 & 2 \\ 3 & -1 \end{vmatrix} \bar{k}$$

$$= 4\bar{i} - 2\bar{j} - 7\bar{k}$$

$$= \langle 4, -2, -7 \rangle \quad \checkmark$$

$$\langle 4, -2, -7 \rangle \cdot \langle 1, 2, 0 \rangle = 0$$

④ Toets:

$$\left. \begin{aligned} (\bar{a} \times \bar{b}) \cdot \bar{a} &= 4 - 4 = 0 \\ (\bar{a} \times \bar{b}) \cdot \bar{b} &= 12 + 2 - 14 = 0 \end{aligned} \right\}$$

② Laat $\vec{a} = \langle 2, 3, -1 \rangle$ en $\vec{b} = \langle -4, -6, 2 \rangle$
Toon aan dat $\vec{a}$ en $\vec{b}$ parallel is.

$$\begin{aligned} \vec{a} \times \vec{b} &= \begin{vmatrix} \vec{i} & \vec{j} & \vec{k} \\ 2 & 3 & -1 \\ -4 & -6 & 2 \end{vmatrix} \\ &= \begin{vmatrix} 3 & -1 \\ -6 & 2 \end{vmatrix} \vec{i} - \begin{vmatrix} 2 & -1 \\ -4 & 2 \end{vmatrix} \vec{j} + \begin{vmatrix} 2 & 3 \\ -4 & -6 \end{vmatrix} \vec{k} \\ &= 0\vec{i} - 0\vec{j} + 0\vec{k} \\ &= \langle 0, 0, 0 \rangle \end{aligned}$$

$\therefore \vec{a}$ en $\vec{b}$ is parallel

③ Bepaal in vektor met lengte 2 wat loodreg is op beide $\vec{a} = \langle 1, -1, 1 \rangle$ en

$$\vec{b} = \langle 3, 2, -1 \rangle$$

- $\vec{a} \times \vec{b}$ • eenheidsvektor $\frac{\vec{a} \times \vec{b}}{|\vec{a} \times \vec{b}|}$
- lengte 2 Dus bepaal $2 \left(\frac{\vec{a} \times \vec{b}}{|\vec{a} \times \vec{b}|} \right)$

$$\vec{a} \times \vec{b} = \begin{vmatrix} \vec{i} & \vec{j} & \vec{k} \\ 1 & -1 & 1 \\ 3 & 2 & -1 \end{vmatrix} = \langle -1, 4, 5 \rangle \checkmark$$

.....

$-\vec{j} \mid \mid$

Toets:

$$\vec{a} \cdot (\vec{a} \times \vec{b}) = 0 \checkmark$$

$$\vec{b} \cdot (\vec{a} \times \vec{b}) = 0 \checkmark$$

$$|\underline{\bar{a}} \times \underline{\bar{b}}| = |\langle -1, 4, 5 \rangle| = \sqrt{(-1)^2 + 4^2 + 5^2} = \sqrt{42}$$

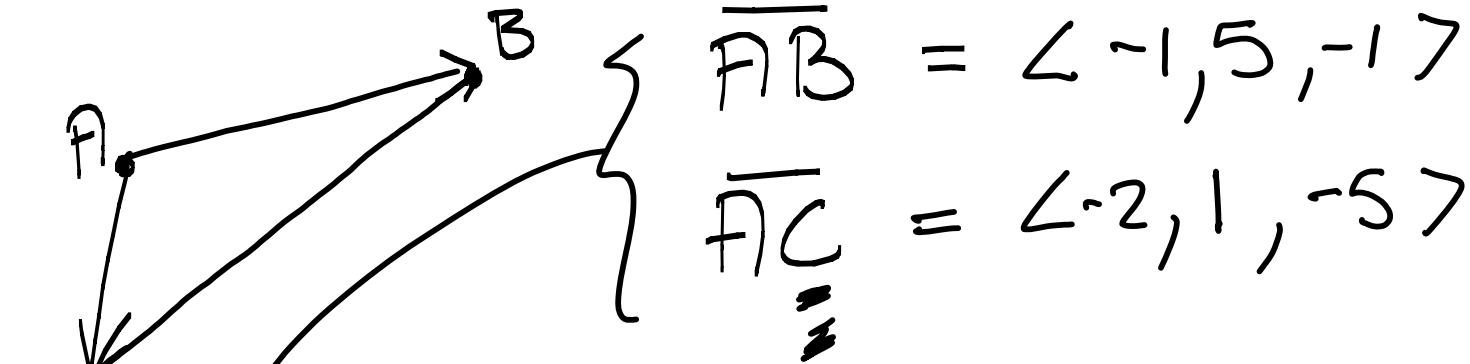
Eenheidsvektor: $\frac{\underline{\bar{a}} \times \underline{\bar{b}}}{|\underline{\bar{a}} \times \underline{\bar{b}}|} = \frac{1}{|\underline{\bar{a}} \times \underline{\bar{b}}|} \langle -1, 4, 5 \rangle = \frac{1}{\sqrt{42}} \langle -1, 4, 5 \rangle$

Die vector $\underline{\bar{b}}$ met lengte 2 is:

$$\underline{\bar{b}} = 2 \times \frac{1}{\sqrt{42}} \langle -1, 4, 5 \rangle = \frac{2}{\sqrt{42}} \langle -1, 4, 5 \rangle$$

④ Bepaal die oppervlakte van die driehoek met toppunte

$$A(2, -3, 4) \quad B(1, 2, 3) \quad C(0, -2, -1)$$



$$\overrightarrow{AB} \times \overrightarrow{AC} = \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ -1 & 5 & -1 \\ -2 & 1 & -5 \end{vmatrix} = \langle -24, -3, 9 \rangle$$

$$\bullet \quad |\overrightarrow{AB} \times \overrightarrow{AC}| = \sqrt{(-24)^2 + (-3)^2 + 9^2} = \sqrt{576 + 9 + 81}$$

$\Downarrow$

$$\text{Opp van } 11 \text{ gram} = \sqrt{576 + 9 + 81}$$

$$\text{Opp van } \Delta = \left(\frac{1}{2}\right) \sqrt{576 + 9 + 81}$$